

CyberDefense Pro 2.0

2.6.4 Diamond Model of Intrusion Analysis

This lesson covers the following topics:

- The Diamond Model of intrusion analysis
- The four categories of the Diamond Model
- Analytic example of the Diamond Model

The Diamond Model of Intrusion Analysis

The Diamond Model of Intrusion Analysis is set out in a paper by Sergio Caltagirone, Andrew Pendergast, and Christopher Betz (activeresponse.org/wp-content/uploads/2013/07/diamond.pdf). The Diamond Model suggests a framework to analyze an intrusion event by exploring the relationships among four core features: adversary, capability, infrastructure, and victim. These four features are represented by the four vertices of a diamond shape. Meta-features may also describe each event, such as date/timestamps, methodologies, resources, direction, kill chain phase, result, etc. Each feature is also assigned a confidence level, indicating data accuracy or the reliability of a conclusion or assumption assigned to the value by analysis. The higher the value the higher the confidence.

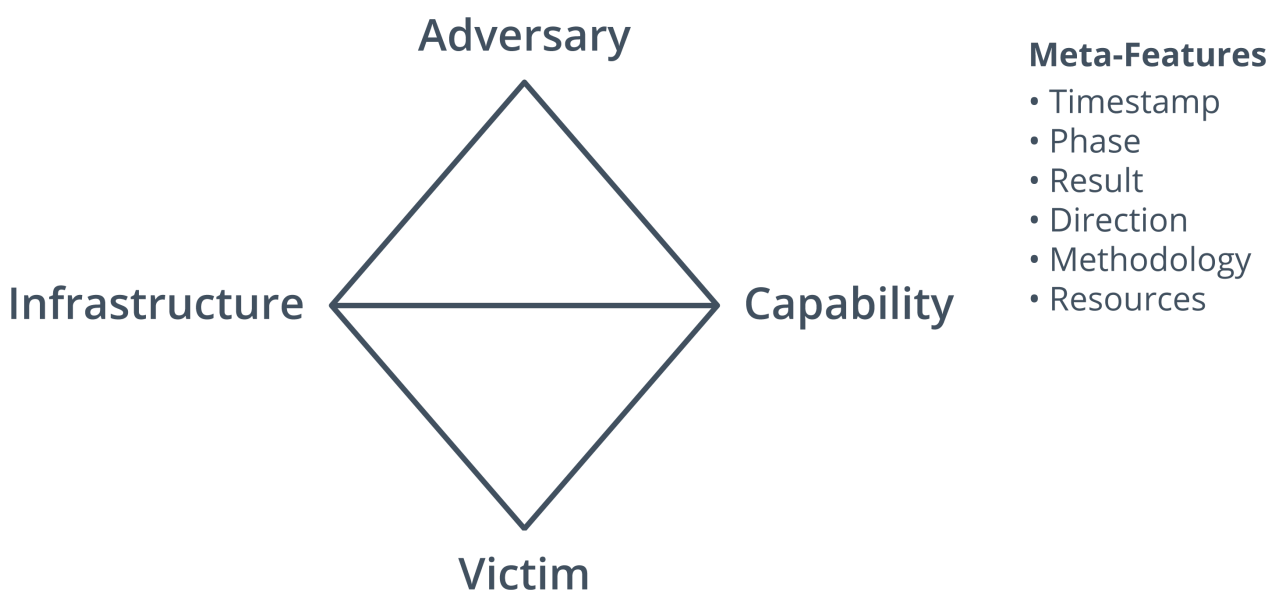


Image Released to public domain by Sergio Caltagirone, Andrew Pendergast, and Christopher Betz [activeresponse.org/wp-content/uploads/2013/07/diamond.pdf].

▼ Description

The image illustrates the Diamond Model of Intrusion Analysis. At the center is a diamond shape oriented with one point up. Each of the diamond's four vertices is labeled to represent a key feature of an intrusion event: "Adversary" (top), "Victim" (bottom), "Infrastructure" (left), and "Capability" (right). To the right of the diamond, a titled list reads "Meta-Features," which includes: Timestamp, Phase, Result, Direction, Methodology, and Resources. These meta-features provide additional descriptive context for an intrusion event, beyond the four primary features captured in the diamond.

The Four Categories of the Diamond Model

The four core features of the Diamond Model are:

Feature	Description
Adversary	This element represents the individual or group responsible for the intrusion. Adversaries can include nation states, criminal organizations, hacktivists, or malicious insiders.
Infrastructure	This element refers to the tools and resources used by the adversary to carry out the intrusion. Tools include malware, exploit kits, command and control servers, and other types of network infrastructure.
Capability	This element describes the technical skills and aptitude of the adversary, such as their ability to craft advanced techniques to evade detection, exploit vulnerabilities, and persist on target systems.
Victim	The victim element represents the organization or individual the adversary has targeted, such as government agencies, businesses, or individuals. Victims vary in size, industry type, and defensive capabilities.

By analyzing these four elements, security analysts can better understand the tactics, techniques, and procedures used by the adversary and develop effective strategies to defend against future intrusions. Additionally, the Diamond Model can identify patterns and trends to help organizations improve their cybersecurity posture.

Analytic Example of the Diamond Model

The power of the model lies in the ability to pivot along the vertices of the diamond to produce a complete analysis and correlation of the indicators of compromise (IoCs) that represent the event. The following is an example of pivoting:

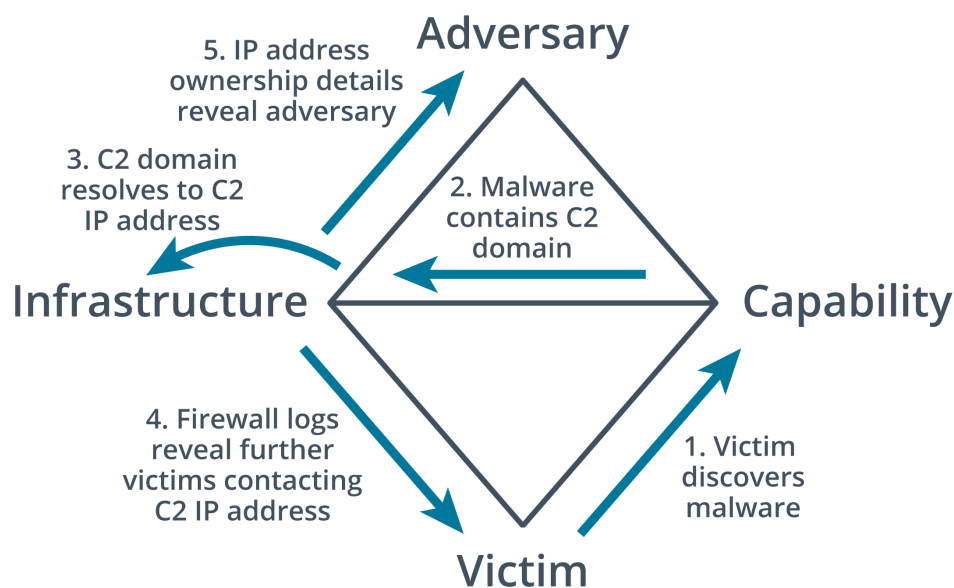


Image released to public domain by Sergio Caltagirone, Andrew Pendergast, and Christopher Betz [activeresponse.org/wp-content/uploads/2013/07/diamond.pdf].

▼ Description

The image depicts the Diamond Model of Intrusion Analysis as a diamond shape with four labeled corners: Adversary (top), Capability (right), Victim (bottom), and Infrastructure (left). Blue arrows illustrate an example analytic pivoting process between the vertices. The step-by-step process includes:

1. Arrow from Victim to Capability: "1. Victim discovers malware."
2. Arrow from Capability to Infrastructure: "2. Malware contains C2 domain."
3. Arrow from Infrastructure to Infrastructure: "3. C2 domain resolves to C2 IP address."
4. Arrow from Infrastructure to Victim: "4. Firewall logs reveal further victims contacting C2 IP address."
5. Arrow from Infrastructure to Adversary: "5. IP address ownership details reveal adversary." These steps demonstrate how analysts can traverse relationships among the four features (adversary, capability, infrastructure, and victim) to uncover a broader understanding of a cyber intrusion event's indicators and context.

Events can be linked into attack graphs and activity threads, graphed along each vertex, representing the paths an adversary could take (if analyzing an attack in progress) and those that

were taken (if analyzing past activity). Also, threads can be assigned to activity groups, which can be used to represent campaigns by particular adversaries.

While the Diamond Model is difficult to apply to manual "pen and paper" analysis, it has been used to develop automated threat intelligence analysis engines which are often integrated with various SIEM platforms.

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